SHARED THIRD-YEAR MULTIMEDIA

Introduction to Multimedia

LIU08106 · BLISS III | RIU08106 · BRIM III

From media concepts to responsible information services.

Key question: How do different media forms help users understand and access information?

What will you learn?

- Define multimedia, name its elements and explain their roles.
- Compare multimedia, hypertext, hypermedia and presentation structures.
- Distinguish analogue from digital media and explain digitisation.
- Evaluate multimedia services in libraries and records/archives environments.

By the end, you should be able to propose a small, purposeful multimedia service.

Why multimedia matters

- Information services must communicate, not merely store files.
- Some ideas are easier to explain with images, sound or motion than text alone.
- Well-designed navigation connects information to the user’s task.
- Media must be accurate, inclusive and maintainable.

Example: a library service guide combines instructions, photographs, an optional video and links.

What is multimedia?

- Multimedia is the purposeful combination of two or more media forms.
- Common forms include text, still images, audio, video and animation.
- Digital multimedia may also include interaction and navigation.
- The goal is clearer communication or improved task completion—not decoration.

A narrated photo exhibition is multimedia; an unrelated decorative picture may add little value.

Element 1: Text and images

- Text supplies precise names, descriptions, instructions and searchable language.
- Images show appearance, evidence, spatial relationships and visual examples.
- Together, a photograph and a reliable caption explain more than either alone.
- Meaningful images may need text alternatives for accessibility.

BLISS: book cover + catalogue description. BRIM: archival photograph + provenance caption.

Element 2: Audio, video and animation

- Audio presents speech, music, environmental sounds and first-person testimony.
- Video shows movement, procedures and events over time.
- Animation illustrates change or relationships without requiring live footage.
- Captions, transcripts and user controls are important delivery considerations.

Examples: an oral-history excerpt, catalogue-search demonstration, animated records life cycle.

Interaction and user control

- A user may click, search, filter, pause, replay or choose a route.
- Interactive features should help the task, not increase unnecessary complexity.
- Labels, feedback and navigation need to be clear and predictable.
- Essential functions should not depend on mouse-only actions.

A searchable digital collection is interactive; a fixed documentary can still be multimedia.

A complete BLISS example

- Problem: new students struggle to locate library services.
- Text: directions and service descriptions. Images: desk and building photos.
- Video: optional catalogue-search demonstration. Links: choose the service.
- Measure success by asking students to find a service without assistance.

Every media choice should solve a specific information problem.

A complete BRIM example

- Problem: visitors need context for historical documents and photographs.
- Images: access copies of selected records. Text: dates and descriptions.
- Audio: authorised testimony with a transcript. Links: timeline and themes.
- Document provenance, permissions, source accuracy and access restrictions.

For records and archives, context and authenticity are as important as attractive presentation.

Characteristics of multimedia systems

- Integration: media elements form a coherent resource.
- Purpose: content and interaction serve a defined audience.
- Production and delivery: capture, editing, storage, interface and access devices.
- Quality: accurate, usable, responsive and sustainable delivery.

A multimedia system is content + technology + people + workflow.

Multimedia vs hypertext vs hypermedia

- Multimedia concerns the types of media combined.
- Hypertext connects text through links.
- Hypermedia links information across multiple media forms.
- A resource can be both multimedia and hypermedia, but the terms are not synonyms.

A video is multimedia. A linked collection of image, audio and text records is hypermedia.

Linear, non-linear and hybrid

- Linear: a mainly predetermined order (e.g., a narrated orientation film).
- Non-linear: users select paths (e.g., a searchable digital exhibition).
- Hybrid: guided sequence plus optional links and branches.
- Choose based on users and task; no single structure is always best.

Think: guided instructions or choose-your-own-route exploration?

Analogue and digital information

- Analogue: information represented through continuously varying physical signals/properties.
- Digital: information encoded as discrete values for storage and processing.
- Analogue examples: cassette tape, photographic film.
- Digital examples: WAV voice recording, TIFF scan, MP4 video.

The digital file is a representation; it does not automatically inherit every property of the original.

How digitisation works

- Select and assess the analogue original.
- Capture using a scanner, camera or audio interface.
- Check the resulting file and document capture settings and source details.
- Store a protected master and prepare suitable access versions when needed.

Scanning alone is not a complete quality-controlled digitisation workflow.

Inside a multimedia system

- Input: camera, scanner, microphone or collected digital asset.
- Processing: image/audio/video editing and authoring software.
- Storage: managed files, metadata and documented versions.
- Output: browser, mobile device, kiosk, projector or repository.

An information professional plans the whole path from source to trustworthy access.

Choose media for a purpose

- Use text when precision, retrieval or quick scanning matter.
- Use images when visual identification or comparison matters.
- Use audio/video when testimony or process over time matters.
- Use links and interaction when users need to search or explore choices.

Avoid adding a video where a clear paragraph or labelled image would work better.

Applications in librarianship

- Library orientation and information-literacy tutorials.
- Digital exhibitions and enriched catalogue resources.
- Reading promotion, subject guides and public education.
- Community-history collections and accessible learning resources.

A successful library resource supports discovery, learning and independent access.

Applications in records and archives

- Access copies of historical records and photographs.
- Oral-history collections and virtual archival exhibitions.
- Records-management awareness and procedure demonstrations.
- Linked finding aids with provenance and restriction information.

Preserve original context and document any modification to access copies.

Benefits: what improves?

- Comprehension: different media can explain different aspects of a topic.
- Engagement: testimony and visual evidence can add context.
- Access: remote users can consult selected resources online.
- Protection: access copies can reduce unnecessary handling of fragile originals.

Benefits are potential outcomes; verify them through actual user tasks.

Challenges: what can go wrong?

- Large files, weak networks, limited devices or skills.
- Inaccessible controls, missing captions or unreadable text.
- Copyright, consent, privacy and restricted-records issues.
- Misleading edits, poor descriptions, broken links and file obsolescence.

A visually appealing product can still be ineffective, inaccessible or untrustworthy.

Accessibility in practice

- Use meaningful headings and readable text contrast.
- Provide captions for relevant video and transcripts for audio.
- Describe informative images and label navigation clearly.
- Let users control playback and access essential content by keyboard.

Accessibility means enabling equivalent information and function for diverse users.

Mini case: choose a design

- Scenario: a low-bandwidth student needs a guide to library borrowing.
- Option A: a large autoplay video with no transcript.
- Option B: short readable steps, two labelled images and an optional small video.
- Explain which option better serves the user and what you would test.

Choice B is usually more robust: essential guidance is available without downloading the video.

A simple production workflow

- 1. Define the problem, audience and purpose.
- 2. Select sources, permissions and appropriate media.
- 3. Plan navigation, then create and integrate assets.
- 4. Test accuracy, accessibility and usability; publish and maintain.

Multimedia quality emerges from a documented workflow, not a software feature.

Review: explain, do not memorise

- What makes a resource multimedia rather than simply digital?
- When is a multimedia resource also hypermedia?
- How would BLISS and BRIM projects use similar media differently?
- What should you consider before publishing a scanned record and an interview?

Next: complete the media explorer, practical audit, review questions and quiz.